

Kevin Maa

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Information Technology with Honors | GPA: 3.62

May 2027

Minor: Secure Comp & Networks | *Honors:* Burnett Honors College – University Honors

Relevant Courses: Computer Architecture Concepts, Network Security & Privacy, Security in Computing

Affiliations: IEEE UCF

PROJECTS

KnightCore – RISC-V SoC – 2026 AMD Open Hardware Competition – Group Project

Feb 2026 – Present

- Serving as RTL & Systems Architect for a heterogeneous robotics control SoC targeting the TUL PYNQ-ZU Zynq UltraScale+ XCZU5EG, designing a custom RV32IM soft-core as the real-time kinematic coprocessor responsible for closed-loop joint control of a 6-DOF robotic arm.
- Implementing a hardware CORDIC coprocessor mapped into the MEM stage and a memory-mapped UART controller with TX/RX FIFO buffering, enabling deterministic trigonometric angle resolution for inverse kinematics and non-blocking serial communication with STS3215 servo motors.
- Designing a tri-BRAM memory architecture with instruction, data, and IPC mailbox blocks wrapped in AXI4-Lite with a combinational address decoder, defining the system memory map as a shared contract to allow all four subteams to develop in parallel without blocking dependencies.

CoreRV – Personal RISC-V SoC – Solo Project

Summer 2025 – Present

- Designing a custom RV32IMA soft-core SoC in SystemVerilog targeting the Red Pitaya Stemplab 125-14 FPGA, with the goal of running Doom bare-metal, built around a 5-stage pipeline with full hazard detection, data forwarding, and load-use stall logic.
- Implementing a multi-cycle iterative hardware divider and 3-cycle pipelined multiplier with dedicated FSMs and pipeline stall integration, alongside a Machine-mode CSR subsystem supporting precise trap handling, Writeback-stage exception delivery, and MRET return-from-trap.
- Actively adding dynamic branch prediction, L1 instruction and data caches, UART, a memory-mapped timer, SPI flash, and a framebuffer with HDMI output to complete the SoC peripheral stack.

ESP32 Motion Gesture Classifier – Solo Project

Mar 2026 – Present

- Building an embedded gesture recognition system on an ESP32 DevKit V1 wired to an MPU-6050 6-axis IMU over I2C, writing C++ firmware to stream raw accelerometer and gyroscope data at 77Hz for data collection.
- Training a deep learning classifier in Edge Impulse using Spectral Analysis and Fast Fourier Transform feature extraction to distinguish four motion classes: idle, wave, circle, and thrust, achieving high accuracy on-device.
- Deploying the trained model as a C++ library back onto the ESP32 for fully offline real-time inference, completing a full TinyML pipeline from raw sensor data to edge deployment without cloud dependency.

4 Cluster PC Home Lab – Solo Project

Sep 2025 – Present

- Designing and deploying distributed computing infrastructure using 4 Mini PCs configured in a Proxmox High Availability cluster with ZFS replication for self-hosted services, network experimentation, and cybersecurity training.
- Implementing containerized services using Docker and Kubernetes for application deployment, alongside network security tools including Wireshark, Security Onion, and Kali Linux for threat detection.
- Configuring network segmentation with VLAN architecture, OPNsense firewall rules, and VPN tunneling through WireGuard to isolate services and implement enterprise-grade security best practices.

LEADERSHIP

IEEE-Eta Kappa Nu Zeta Chi – Outreach Officer

Feb 2026 – Present

Electrical and Computer Engineering Honor Society, University of Central Florida

- Inducted into HKN as one of the top-performing Information Technology students meeting the chapter GPA threshold, joining a selective cohort of ECE and CPE majors recognized for academic excellence.
- Serving as the sole Outreach Officer, leading industry outreach efforts by identifying and contacting technology and computing companies to establish collaborations and arrange virtual tech talks for chapter members.
- Drafting and sending professional correspondence to engineering and AI organizations across computing, semiconductor, and research industries to build relationships between the chapter and industry partners.

TECHNICAL SKILLS

Languages: SystemVerilog, C, C++, Python, Bash, Assembly (RISC-V), TypeScript, JavaScript, HTML, CSS

RTL and Hardware Design: RISC-V ISA, Pipelined CPU Design, AXI Interconnect, Memory-Mapped I/O, BRAM, CORDIC, UART, Hazard Detection and Forwarding

Embedded and Edge AI: ESP32, Raspberry Pi, MPU-6050, I2C, Edge Impulse, TinyML

Virtualization and Infrastructure: Proxmox, ZFS Replication, High Availability Clustering, Network Segmentation

Network Security: Wireshark, Security Onion, Kali Linux, OPNsense, WireGuard VPN

Tools: Git, GitHub, Linux, Windows, VSCode, Vivado, Docker, Kubernetes, VLAN